

Grade 8
Unit 2
Lesson 4 Practice

Name Answer Key
Date _____
Period _____

1. Find the square root of each of the following.

$$\sqrt{81} = 9$$

$$\sqrt{-81} = \text{No real solution}$$

2. Find the cube root of each.

$$\sqrt[3]{64} = 4$$

$$\sqrt[3]{-64} = -4$$

3. Why is there a non-real root for $\sqrt{-81}$, however there is a real root for the $\sqrt[3]{-64}$?
There is no real root that when squared will give -81 . $\sqrt[3]{-64}$ has a real root because $(-4)(-4)(-4) = -64$.

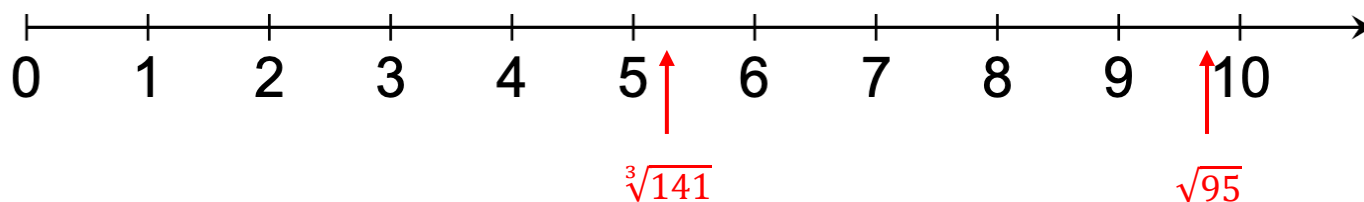
Consider $\sqrt{95}$, an irrational number.

4. Between which two perfect squares does $\sqrt{95}$ lie?
81 and 100
5. Between which two integers is the value of $\sqrt{95}$.
9 and 10

Now consider $\sqrt[3]{141}$, another irrational number.

6. Between which two perfect cubes does $\sqrt[3]{141}$ lie?
125 and 216
7. Between which two integers is the value of $\sqrt[3]{141}$.
5 and 6

8. Plot and label the approximation of $\sqrt{95}$ and $\sqrt[3]{141}$ on the number line.



9. Lauren was trying to approximate the solution to $n^2 = 115$. She said that it lies between $\sqrt{100}$ and $\sqrt{121}$. Therefore, it lies between 4 and 5, however a little closer to 5. She approximated it to be about 4.7. Do you agree or disagree with her reasoning? Explain.

No. $\sqrt{100} = 10$ and $\sqrt{121} = 11$. The approximated value should lie between 10 and 11, closer to 11, or about 10.7.

10. Approximate the solution to $n^2 = 115$ and then plot value on the number line below.

